

STEVE BRONDER

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PROFILE

- Quantitative Developer: Computational Mathematics and High Performance Computing
- Large body of open source contributions, focused on statistical modeling and optimizations for modern CPU and GPU architectures.
- Track record of thoughtful software design that optimize performance for the end-users workflow. I maintain and develop one of the fastest **C++ math libraries** for automatic differentiation.

EXPERIENCE

Flatiron Institute

Software Developer

Apr 2023 – Present

New York, NY

- Wrote the backend for the **Pathfinder** algorithm in Stan for parallelizable variational inference (VI). The Algorithm can scale up to a million parameters and shows a 3x improvement relative to other VI methods.
- Authored **riccaticpp**, which is 5x faster than any other solver for partially oscillatory second order linear ODEs.
- Wrote the Embedded Laplace approximation inside Stan's NUTS Sampler. This method allows for methods normally requiring millions of parameters to be estimated with 10 fold fewer summary statistics.

Columbia University

Computational Mathematics Consultant

Feb 2020 – May 2022

New York, NY

- Led a four person team designing a new struct of arrays memory layout for automatic differentiation for linear algebra. The new pattern is much more cache friendly and allows the use of SIMD intrinsics. Using the new memory layout cuts runtimes for Stan models by **20–60%**.
- Added native multi-threading to run multiple chains in Stan's MCMC API, trimming peak RAM per job linearly by the number of cores through shared memory across cores.
- Built a C++14 legacy requires type traits library adopted by the Stan Math library. This allows for safe perfect forwarding for functions over specific types.
- Setup tooling for profiling Stan models and math functions with **perf** and flamegraphs; pinpointed cache-miss hot spots and false sharing in reduce sum problems.

Capital One Commercial Bank – FIG

Principal Data Scientist

Apr 2017 – Jan 2020

New York, NY

- Developed probability of default models for leveraged-loan and CDO/CLO portfolios to estimate worse case collateral pool outcomes.
- Built ML classifier in R for predicting Moody's ratings for unrated collateral with **3x lower** MAE than legacy scorecards.
- Automated data validation pipeline (Python + asyncpg) for collateral Excel files sent from the CDO servicer. saved 6 FTE-days/month in manual reconciliation.

SELECTED OPEN-SOURCE PROJECTS

- **Stan Math** – C++17 automatic differentiation library. Used as the backend for the Stan programming language. Many benchmarks show this is the fastest CPU Automatic Differentiation library.
- **riccaticpp** – Fast solver for stiff/oscillatory ODEs; benchmarks show **5x** speedup vs. state-of-the-art.
- **StanMathCL** – Asynchronous OpenCL backend for Automatic Differentiation; first GPU Cholesky for reverse-mode AD; **8x** speed-ups on 10kx10k matrices via kernel fusion.
- **Eigen** – Many patches and assisted Eigen team in upgrading to C++14.
- **quantmod** – Built API connections allowing access to Tiingo for stock data and Orats for options data.
- **mlrfe** – R package merging ML hyper-parameter tuning with classical forecasting; presented at R/Finance 2017.

TECHNICAL SKILLS

Languages: C++17/20, Python, R, Stan, CUDA, OCaml, SQL

Tools: Git, Docker, CMake, perf, magic-trace, flamegraphs, Linux, Postgres, clang-tidy/format, AWS

EDUCATION

Columbia University – M.A. Applied Statistics 2016

Duquesne University – B.S.B.A. Economics & Mathematics 2014